

BCA II SEMESTER
MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE (BCA-108)
ASSIGNMENT-1

1. If the mean of 7, 9, 11, x and 15 is 12. Find the value of x.
2. Find the median for the series: 10, 12, 8, 9, 70, 60, 40, 80.
3. Define mode.
4. Define measure of dispersion. List the method of measure of dispersion.
5. Find the standard deviation of 11, 14, 15, 17, 18.
6. Explain the following briefly with suitable examples:
 - a) Variance and its calculation and uses.
 - b) Presentation of frequency distribution table and its uses.
7. The following table shows the marks secured by 100 students in an examination. Find the mean marks obtained by the students.

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	15	20	35	20	10

8. Find the mean for the following data:

Class Interval	10-19	20-29	30-39	40-49	50-59	60-69
No. of Students	5	12	17	20	18	8

9. In a study on patients, the following data were obtained. Calculate the arithmetic mean.

Age in Year	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89
No. of Cases	1	0	1	10	17	38	9	3

10. Find the missing frequencies in the following frequency distribution table, it being given that the mean of this distribution is 50:

Class Interval	0-20	20-40	40-60	60-80	80-100	Total
Frequency	17	?	32	?	19	120

11. Find the mean and median for the following data:

C.I.	200-240	240-280	280-320	320-360	360-400	400-440	440-480
Frequency	1	0	1	10	17	38	9

12. Find the median of the following frequency distribution:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	15	17	19	27	21	15

13. Find the median of the following frequency distribution:

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	15	17	19	27	19	12

14. Calculate the standard deviation of the following distribution.

Marks	15	20	25	30	35	40	45	50
No. of Students	3	4	5	3	3	2	3	2

15. Calculate the standard deviation of the following distribution.

Age in Year	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Cases	3	61	132	153	140	51	2

16. The mean and variance of 7 observations are 8 and 16 respectively. if five of the observations are 2, 4, 10, 12, 14, find the remaining two observations.